

Name: Justin M. Iadicola

Date: 5/11/2026

Class: 8

MyDesign Portfolio

Element C

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The system must use RFID technology to uniquely identify students within 2 seconds of scanning their RFID tag.

This requirement directly connects to research from Element B regarding RFID attendance systems, which demonstrated that RFID technology improves identification accuracy and reduces manual tracking errors. Teachers identified student verification as one of the largest weaknesses of traditional paper pass systems because students could misuse passes or remain unaccounted for.

The design requirement was approved by the stakeholder. Teachers stated that quick and accurate identification is necessary because classrooms cannot afford delays during instructional time. No revisions were required.

Design Requirement #2 (HIGH Priority)

The program must record the student's destination and automatically track the amount of time the student spends outside of class.

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Research from Element A showed that schools struggle with lost instructional time and lack of accountability when students leave classrooms. Electronic hall pass systems such as SmartPass demonstrated that automated timing systems improve hallway management and reduce excessive time outside class.

The design requirement was approved by the stakeholder. Teachers emphasized that time tracking is important because students frequently exceed reasonable pass durations without teacher awareness.

Design Requirement #3 (HIGH Priority)

The MATLAB interface must allow users to select destinations from a predefined list while also supporting a custom "Other" destination option.

Peer feedback from the Phase 3 checkpoint suggested expanding the list of available destinations and adding flexibility for special circumstances. This requirement ensures the system can support common school destinations while remaining adaptable to emergencies or uncommon requests.

The design requirement was approved by the stakeholder. Students and teachers agreed that a flexible destination system would make the program more practical for real classroom situations.

Design Requirement #4 (HIGH Priority)

The system must allow multiple students to be registered in the database while



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maintaining unique student IDs between 6 and 8 numeric digits.

Research from Element A identified accountability and organization as major concerns for schools. Using unique student IDs reduces confusion between students and improves record accuracy when multiple users are stored in the system.

The design requirement was approved by the stakeholder. Teachers stated that student identification numbers are necessary to differentiate students with similar names and maintain accurate records.

Design Requirement #5 (HIGH Priority)

The system must display whether a student returned "On Time" or "Late" based on a teacher-defined time limit.

Current hall pass systems often fail to provide immediate feedback about student tardiness. This requirement directly addresses the original problem identified in Element A involving students taking excessive time outside the classroom.

The design requirement was approved by the stakeholder. Teachers stated that automated late detection would improve accountability and reduce the need for manual monitoring.

Design Requirement #6 (LOW Priority)



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The system should save pass records into a digital log file for future review by teachers or administrators.

Digital record keeping was identified during research as an advantage of electronic hall pass systems because it allows schools to review movement history and identify patterns involving excessive pass usage.

The design requirement was approved by the stakeholder. Administrators believed that stored records could help support disciplinary investigations and attendance reviews.

Design Requirement #7 (LOW Priority)

The program should include a graphical user interface with buttons and menus to improve ease of use for teachers.

Research into existing digital systems showed that user-friendly interfaces improve adoption rates and reduce training time for staff. Teachers indicated that simple controls are important because classroom technology must be efficient and intuitive.

The design requirement was approved by the stakeholder. Teachers stated that a simple graphical interface would make the system easier to use during active classroom instruction.

